

RTD on S32G A53 Release Note

1. Read Me First

RTD on S32G A53 project’s target is to utilize spare A53 core resources to expand real-time computing power. Customers using S32G3/S32G2 can be intended users of this release.

Note: The delivered software does not contain functions for cycle/test recognition or any other optimizations for test situations.

1.1. Requirements

1.1.1. System requirements

See RTD_ON_S32G_A53_UM

1.1.2. Used Documentation

- 1) S32G2RM
- 2) S32G3RM

1.2. Problem Reporting Instructions

S32G Internal Community

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2. Release Description

2.1. Release content

Workspaces for the PFE, IPCF, Bootloader and FreeRTOS integration demo.

The release also contains the RTD on S32G A53 User Manual and this Release Notes file available separately in Flexera.

2.2. What's new

A FreeRTOS project based on the A53 core, integrating LLCE, PFE, IPCF, and BSP.

2.3. Known issues/limitations

Component	Description	Workaround
Bootloader	The current Bootloader's EB interface does not support configuration options for enabling DDR; BL_CONFIGURE_DDR is disabled by default.	After code generation, you need to manually change BL_CONFIGUR to STD_ON.
LLCE	RTD version 5.0.0 has a bug related to LLCE memory; using the default RTD 5.0.0 will generate incorrect Shared_Memory addresses during compilation.	Replace Can_43_LLCE_MemMap.h in RTD 5.0.0 with the version from RTD 5.0.0 QLP03
PFE	Because RTD does not support PFE configuration under Aarch64, some default generated configurations are still Aarch32.	Please follow the instructions from the User Manual for more details on how to accomplish that.
PFE	Due to the higher clock speed of the A53, the initialization of the PFE Slave occurs before the PFE Master initializes the bridge, resulting in the failure of RPC communication for the Slave	Add a wait flag to the PFE Master; the Slave will wait for the Master to complete its initialization before performing its own initialization.
IPCF	The IPCF MCAL code does not adapt to the interrupt numbers of different architectures and still uses the M7 interrupt number to handle A53 interrupts.	Please follow the User Manual for more details on how to accomplish that.

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